

Chapter 8, Problem 7P

Problem

A series RLC circuit has $R = 20\text{ k}\Omega$, $L = 0.2\text{ mH}$, and $C = 5\text{ }\mu\text{F}$. What type of damping is exhibited by the circuit?

Step-by-step solution

Step 1 of 2

Write the expression for neper frequency (α).

$$\alpha = \frac{R}{2L} \dots\dots (1)$$

Substitute $20\text{ k}\Omega$ for R and 0.2 mH for L in equation (1)

$$\begin{aligned}\alpha &= \frac{R}{2L} \\ &= \frac{20 \times 10^3}{(2)(0.2 \times 10^{-3})} \\ &= 50 \times 10^6 \text{ rad/s}\end{aligned}$$

Step 2 of 2

Write the expression for resonant frequency (ω_0).

$$\omega_0 = \frac{1}{\sqrt{LC}} \dots\dots (2)$$

Substitute 0.2 mH for L and $5\text{ }\mu\text{F}$ for C in equation (2)

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(0.2 \times 10^{-3})(5 \times 10^{-6})}} \\ &= \frac{1}{3.16 \times 10^{-5}} \\ &= 31622.7 \text{ rad/s}\end{aligned}$$

From the values of α and ω_0 , it is clear that $\alpha > \omega_0$. Therefore, the response is over damped.

Hence the damping exhibited by the circuit is **over damping**.